

A vertical image on the left side of the slide shows a variety of fresh vegetables, including yellow and red bell peppers, tomatoes, green beans, carrots, and an artichoke, arranged on a rustic wooden surface. A white tablet is partially visible in the bottom left corner, displaying a grey screen.

Similarities and Differences Between Academic and Industrial Environments

Isuru Mohottala : Software Architect

12th Feb 2026



SYSCO

- Sysco Corporation is (NYSE: SYO) the largest food services company in world.
- Sysco was founded in 1969 by John Baugh and is headquartered in Houston, Texas.
- Ranked #56 in 2025 Fortune 500 list.
- \$81.4 billion in revenue in FY25.

A Golden Rule

SOFTWARE ENGINEERING GOLDEN RULE

- Bring me a Pancake



Rule : Ask, Don't Assume



Agenda

- 1 **Thought Leadership**
- 2 **Teams and Process**
- 3 **Presentations and Documentation**
- 4 **Security**
- 5 **AI & Productivity**
- 5 **Learning**

12th Feb 2026

Thought Leadership

TASK TAKERS TO PROBLEM SOLVERS



Class Room



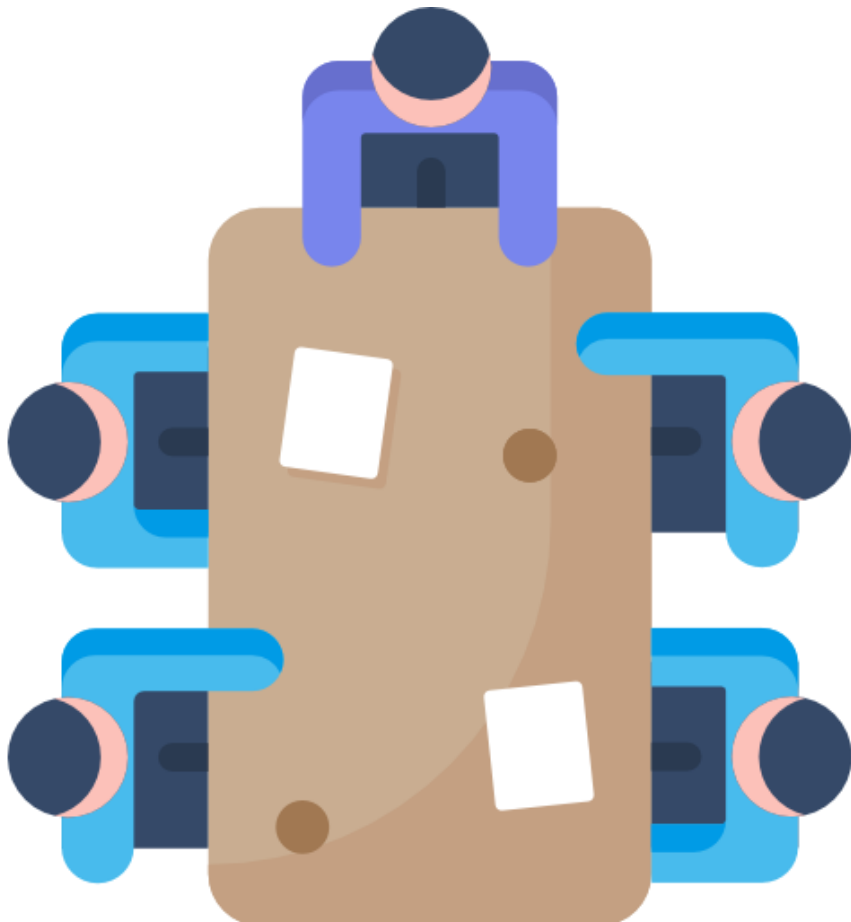
Passengers

How Do I Build This ?

Listening

Agreement

Passive



Board Room

Should We Build This ?

Ownership

Active

Constructive Disagreement



Drivers

Teams and Process

GROUP PROJECTS TO ENTERPRISE APPLICATIONS

University Project : Friends Working for Common Goal

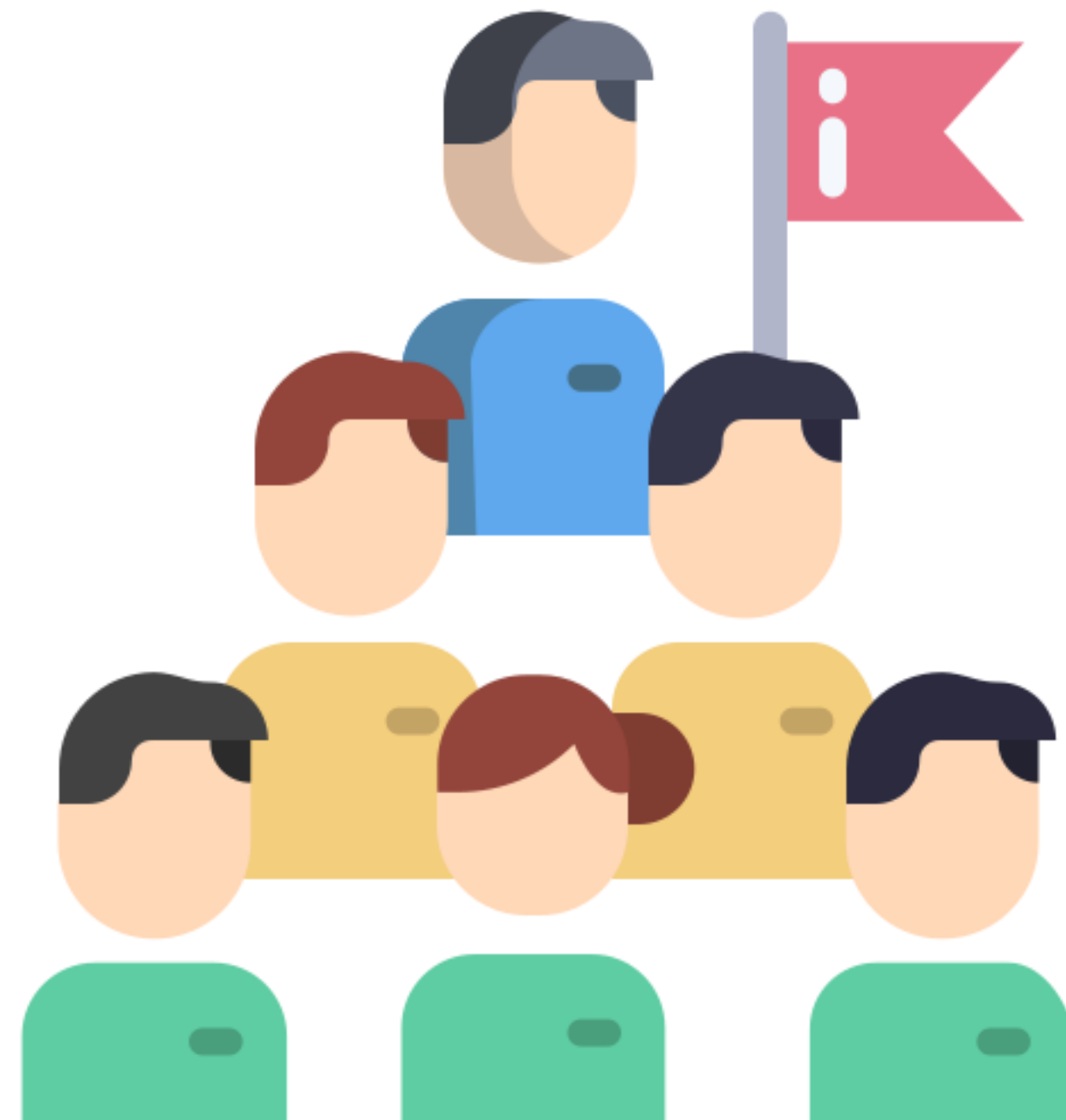


- Collaboration and Team Work
- Clear Communication
- Iterative Refinement
- Continuous Improvement
- Learning Centric

Teams and Process

GROUP PROJECTS TO ENTERPRISE APPLICATIONS

Project Team : Colleagues Working for Common Goal



- Collaboration and Team Work
- Clear Communication
- Iterative Refinement
- Continuous Improvement
- ~~Learning Centric~~
- Defined Roles
- Definition of Done
- Peer Review
- Business Value Centric

Teams and Process

TASK TAKERS TO PROBLEM SOLVERS

University Project

- Death Date : Done after submission.
 - **Building for a grade**
- Definition of Done : Pass the testcases at the Demo
 - **Development 80%**
- Learning a Concept
 - **Writing from the scratch**
- Artificial Constraints
 - **No external libraries / No frameworks**
- Grades
 - **Loose Marks**

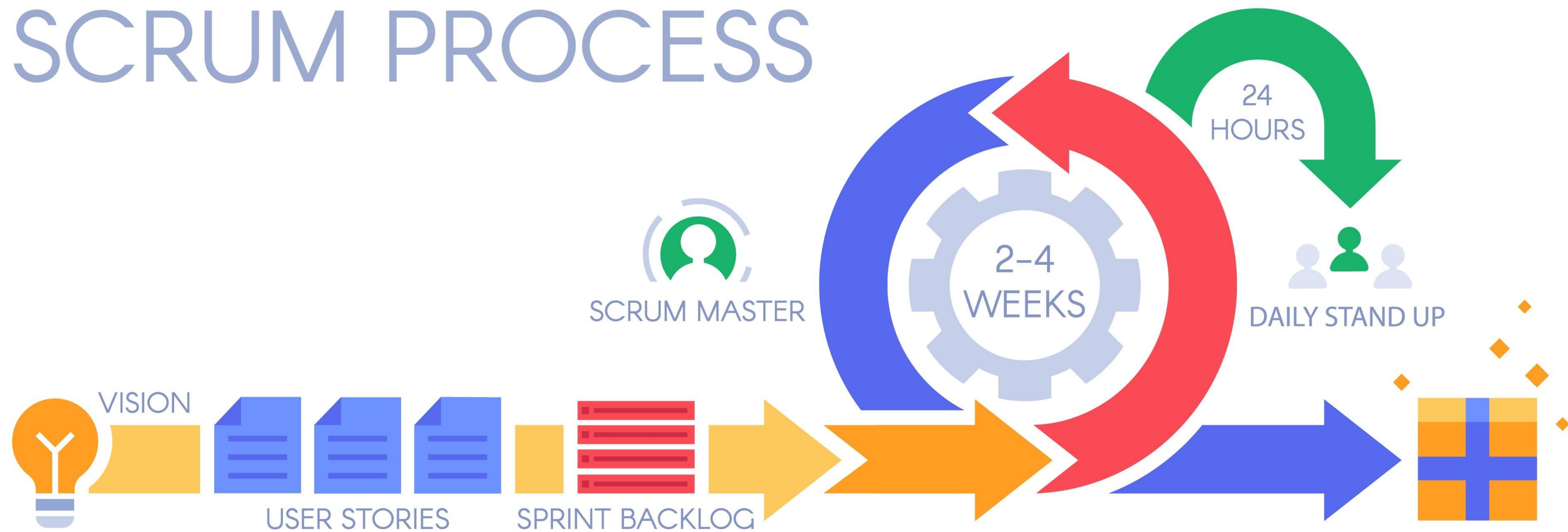
Project Team

- Living Organisms : Maintaince
 - **Support call at 2:00 am**
- Definition of Done : reviwed, tested, deployed to Prod
 - **Development 30%**
- Provide Business Value
 - **Reusability**
- Real Constraints
 - **Budget , Legacy Database**
- Revenue
 - **Loose Money**

Teams and Process

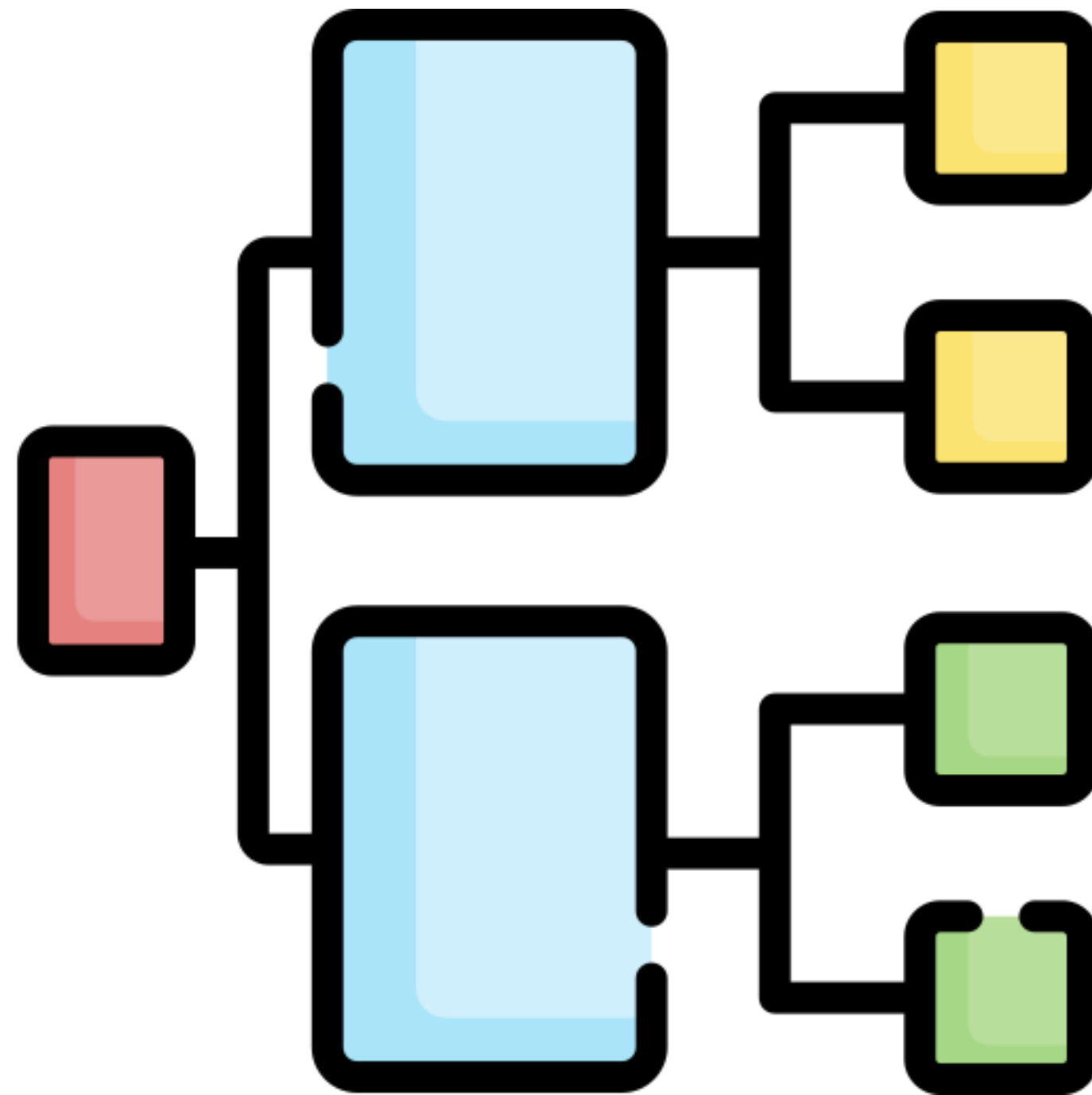
GROUP PROJECTS TO ENTERPRICE APPLICATIONS

SCRUM PROCESS



Presentation, Design, and Documentation

ARCHITECTURAL DIAGRAMS



- Documentation is key aspect in Industry and Academic Environments
- Elevator Pitch : How you present your work
- **Detailed Diagrams to High level view**

UML

Flow Charts

Use Case

Sequence

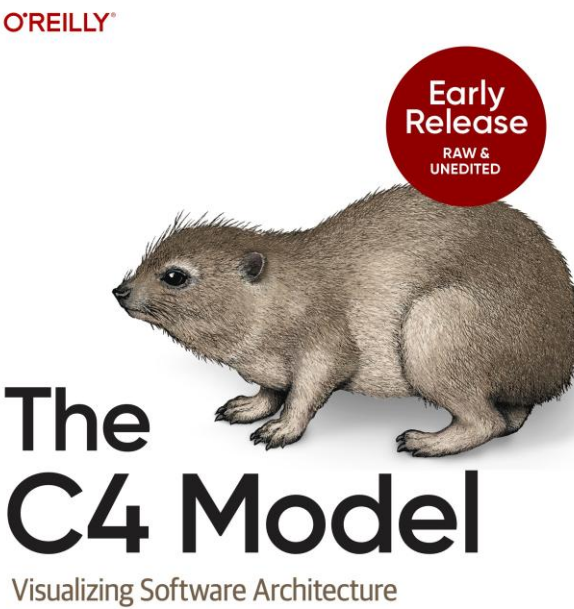
Class Diagrams

Activity Diagrams

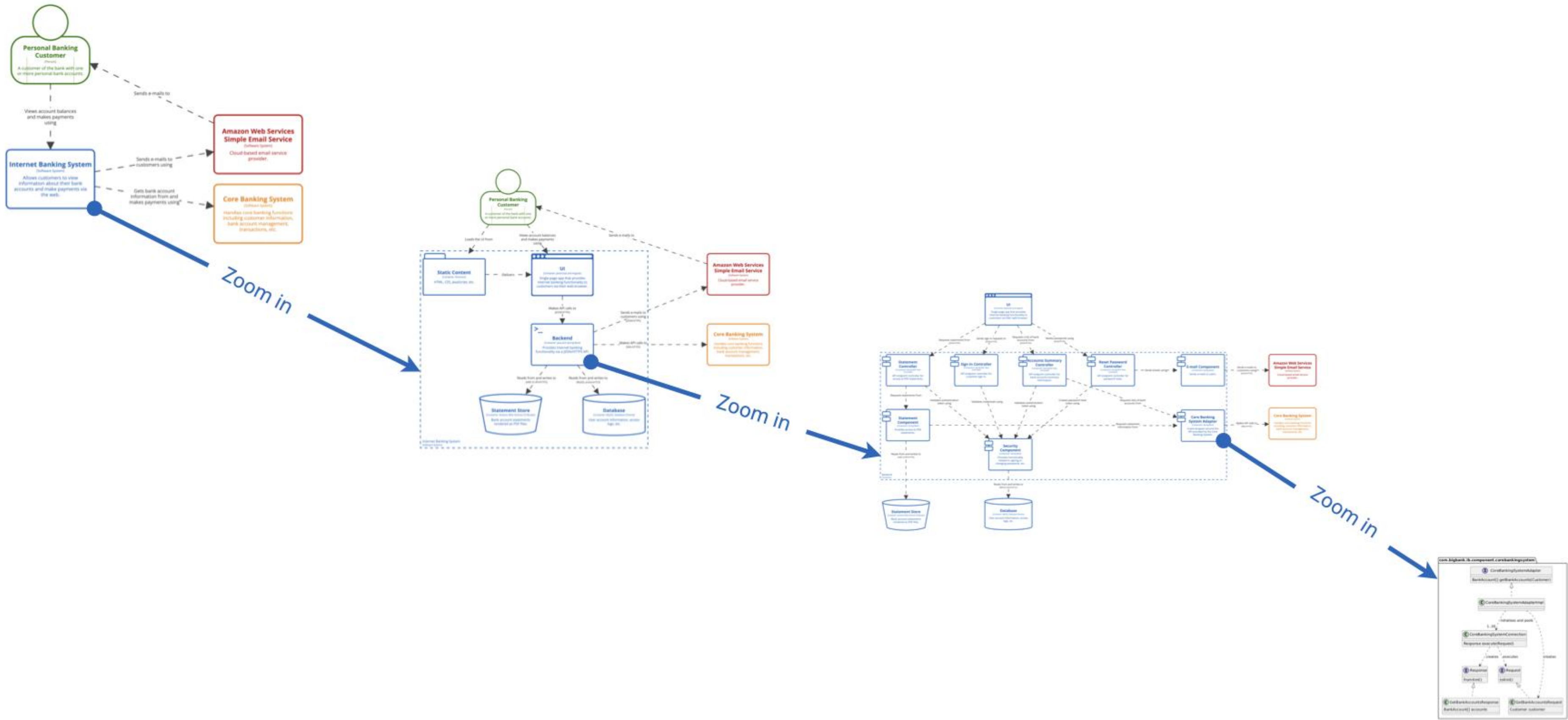
ER Diagrams

C4 Diagrams

ZOOM IN FOR MORE



Simon Brown

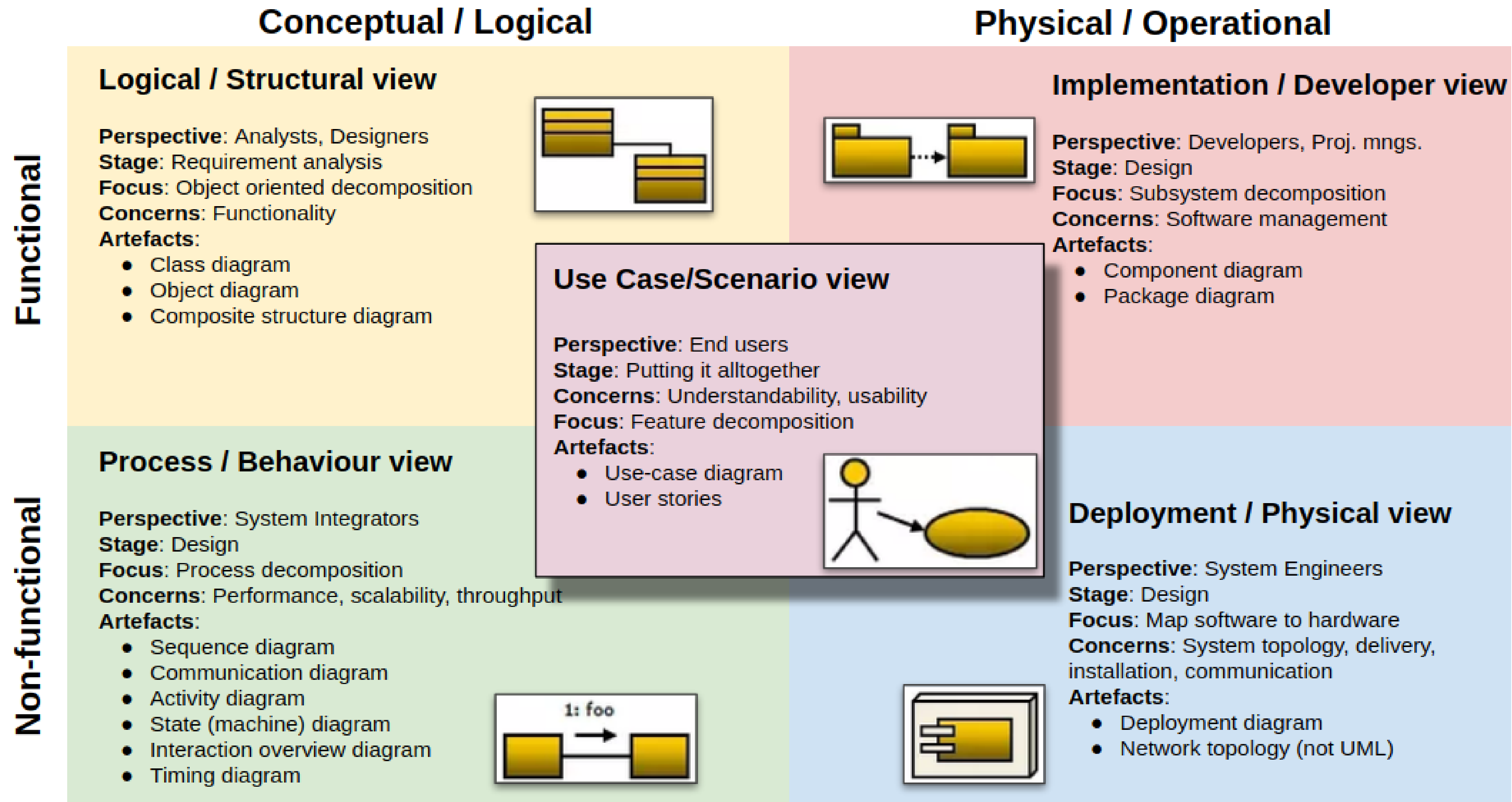


Static structure diagrams

System Context Containers Components Code

4 + 1 View

DISTINCT CONCERNS OF DIFFERENT STAKEHOLDERS



Cyber Security

STANDALONE MODULE TO CONTINUES CONSTRAINTS

University Project

- Module and a Check Point in Final Project
- Open : Any Device, Any Wifi, Any Tool
- Focus is on Hacking the Code
- Simple Passwords on Text files or Hard Coded
- Just Security

Enterprise Application

- “**Shift Left**” Philosophy
- **Zero Trust** : Never trust, always verify
- Social Engineering : **Hacking the Human**
- Secret Management : ‘HashiCorp Vault’, ‘AWS Secrets Manager’, or ‘GCP Secret Manager’
- Security and Compliance : GDPR, SOC2, HIPAA, PCI-DSS



In industry, security is not a checkbox. It is a **Non-Functional Requirement (NFR)** that is just as important as the code actually working. A perfectly functional application that leaks user data is a failure

AI and Productivity

TALK OF THE TOWN

University

Industry

- Short Cut
- Force Multiplier
- Boilerplate Acceleration
- The Search Shift
- Documentation
- The Maintenance Burden
- Security : Secret Leakage
- AI as a Mentor



Copilot



Google Antigravity

Continuous Learning

ALWAYS LEARNING

University

- Studying for the Next Exam

- Research and Reading
- Peer to Peer Learning
- Zero to Hero (Beginner Mind Set)

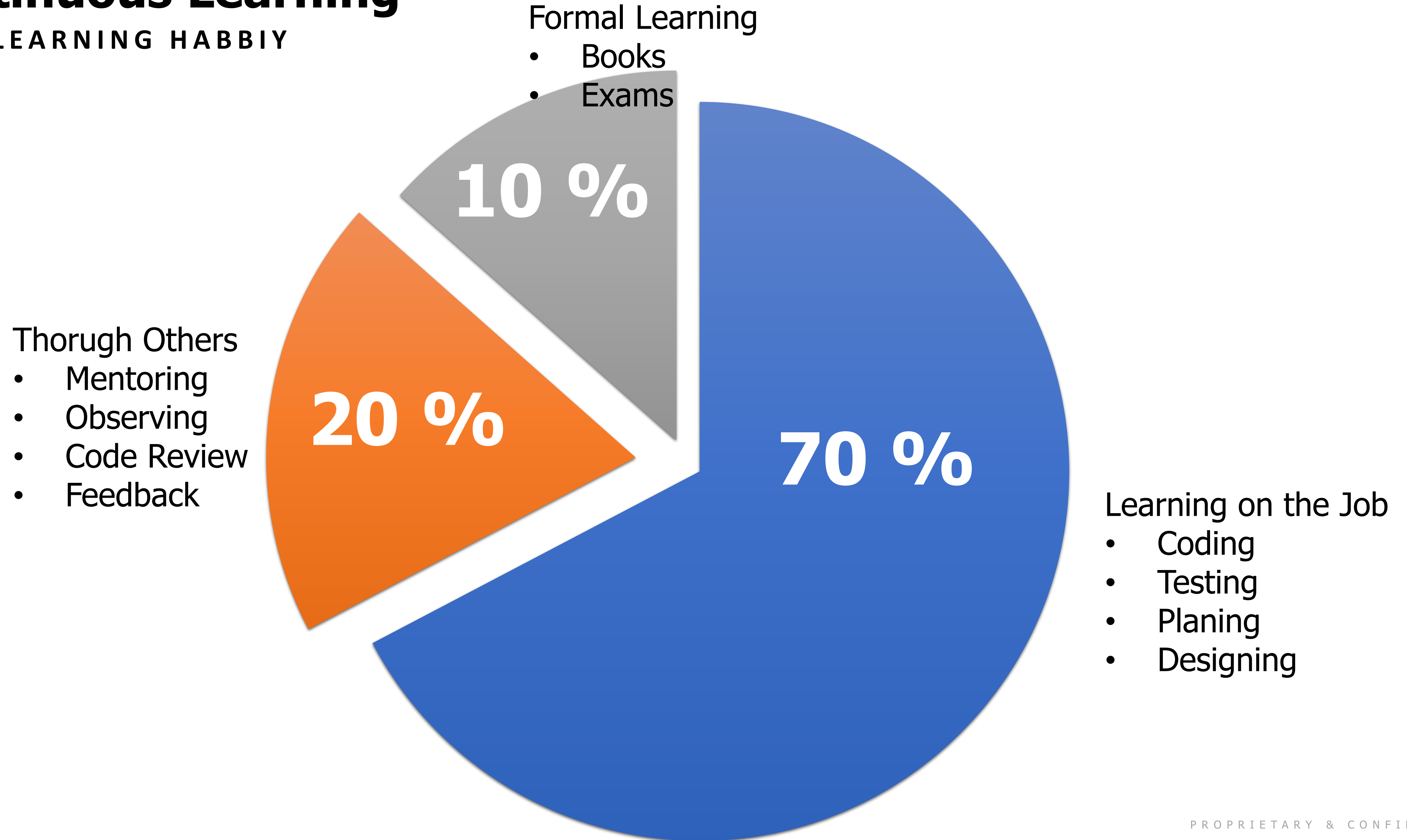
Industry

- Study to Survive
- Half Life Knowledge

In school, you learn to pass. In industry, you learn to **sustain**. The goal isn't to be a walking encyclopedia; it's to be an adaptable problem-solver who knows where to find the answers.

Continuous Learning

KEEP LEARNING HABBIY



Sysco® LABS

Sysco FORTUNE
500
RANK 56™

THANK
YOU!

Sysco® LABS

